

1 Prompt first pass

You are extracting structured information from research papers for a systematic mapping study on data-visiting architectures. I attached 10 documents. Do the following for all these 10 documents.

Definition

Data visiting refers to architectures where computation is brought to the data rather than moving data to a central location.

Important context

All papers are already relevant to data visiting concepts. Your task is NOT to judge inclusion, but to extract comparable architectural features for later selection.

Rules

- Use ONLY information present in the paper. - If unknown, write "unknown". - Be precise and conservative. - Do not decide relevance categories.

Output format (strict JSON only)

```
"title": "", "year": "",
  "architecture_family": "", "architecture_subfamily": "",
  "domain": "",
  "deployment_model": "", "communication_pattern": "", "data_visiting_mechanism": "",
  "key_components": [],
  "architectural_description": "",
  "architectural_detail_score": 0, "data_visiting_strength_score": 0,
  "diversity_signature": ""
```

Scoring guidance

- architectural detail score (0–100): how well architecture is described (components, flows, topology). With this take only network-level architecture, this how (types of) devices communicate and relate to another.
- data visiting strength score (0–100): how strongly computation is moved to data sources
- diversity signature: a short normalized label capturing architecture type, e.g.: "federated_learning_cross_silo" "edge_cloud_hybrid" "data_space_federation" "trusted_research_environment"

2 Prompt second pass

Categorize/classify the articles based on their relevance to a systematic mapping study on data visiting architectures. Use the provided metadata of the articles.

Only assign **High relevance** sparingly, with a target of approximately **30-50 articles in total**.

Definition of data visiting

Data visiting refers to architectures where computation is brought to the data rather than moving data to a central location.

Important context

Assume that all candidate articles have already been screened for relevance to data visiting and architectural concepts. The purpose of this classification is therefore not to determine inclusion or exclusion, but to identify the most useful subset of articles for detailed analysis in the systematic mapping study.

Process

If something does not work or does not make sense, or you cannot access the full text, clarify with me instead of trying to accommodate/figure it out. This does not have to be done fast, but well. I want the results to be reproducible, not easy or fast.

Classification categories

- **High relevance:** The article represents an architecture type that is valuable for the mapping study and provides substantial architectural detail, including components, interactions, communication patterns, deployment topology, or network-level design.
- **Medium relevance:** The article discusses a data-visiting architecture and provides some architectural information, but is either less detailed or highly similar to other articles already represented in the High relevance category.
- **Low relevance:** The article discusses data visiting concepts but focuses primarily on algorithms, applications, governance, performance, or outcomes, with limited architectural detail.

Prioritization criteria

When assigning relevance categories, use the following priorities in order:

1. **Architectural diversity (highest priority)** Prefer articles that represent distinct architecture families, paradigms, deployment models, or communication patterns. Avoid selecting many highly similar articles from the same architecture family.
2. **Architectural detail (second priority)** Prefer articles that clearly describe:
 - System components
 - Interactions between components
 - Communication paths and message flows
 - Deployment topology
 - Network-level architecture
 - Which devices, nodes, services, organizations, or systems communicate with one another
3. **Strength of data-visiting implementation (third priority)** Use this criterion only as a tie-breaker when the first two criteria are similar.

High relevance selection guidelines

- High relevance should contain a broad and representative mix of architecture types.
- Do not allow a single architecture family (e.g., federated learning) to dominate the High relevance category.
- If multiple articles describe highly similar architectures, assign High relevance only to the most architecturally descriptive and representative examples.
- Prefer articles that contain architecture diagrams, component overviews, deployment descriptions, communication flows, or detailed system explanations.

Output

For each article provide:

- Relevance category (High, Medium, or Low)
- Confidence (High, Medium, or Low)
- A brief justification focused on architectural diversity and architectural detail
- The architecture family or architecture type represented by the article